Heart of Algebra

ANSWER KEY

Section 1: Multi-Step Linear Equations & Identity Conditions

1 If $5(2x - 1) - 3(x + 4) = 4(x - 3) + 7$, what is the value of x?

a. 4

b. 6

c. 8

d. 12 ✓

Expand both sides: $10x - 5 - 3x - 12 = 4x - 12 + 7 \Rightarrow 7x - 17 = 4x - 5$. Subtract $4x$ and add 17: $3x = 12 \Rightarrow x = 4$.

2 For what value of a will the equation $3(ax - 2) + 14 = 12x + 8$ have infinitely many solutions for all real numbers x?

a. 2

b. 4 ✓

c. 6

d. 8

Expand: $3ax - 6 + 14 = 12x + 8 \Rightarrow 3ax + 8 = 12x + 8$. Matching coefficients of x: $3a = 12 \Rightarrow a = 4$.

3 If $\frac{2x - 5}{3} = \frac{x + 4}{2}$, what is the value of $3x + 1$?

a. 22

b. 64

c. 67 ✓

d. 70

Cross multiply: $2(2x - 5) = 3(x + 4) \Rightarrow 4x - 10 = 3x + 12 \Rightarrow x = 22$. Then $3x + 1 = 3(22) + 1 = 67$.

Section 2: Linear Inequalities & Feasible Regions

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4 Solve the compound inequality for x: $-5 < 2x + 3 \leq 13$.

a. $-4 < x \leq 5$ ✓

b. $-4 \leq x < 5$

c. $-1 < x \leq 8$

d. $x > -4$

Subtract 3: $-8 < 2x \leq 10$. Divide by 2: $-4 < x \leq 5$.

5 Which of the following points (x, y) is a solution to the system $y > 2x - 3$ and $y \leq -x + 4$?

a. $(1, 2)$ ✓

b. $(3, 5)$

c. $(0, -4)$

d. $(2, 4)$

Test $(1, 2)$: $2 > 2(1) - 3 = -1$ (true), and $2 \leq -(1) + 4 = 3$ (true). Both conditions are satisfied.

6 If $-3x + 7 \geq 22$, what is the greatest possible value of x?

a. -5 ✓

b. -3

c. 5

d. 15

Subtract 7: $-3x \geq 15$. Divide by -3 (reverse inequality sign): $x \leq -5$. The maximum value is -5.

Section 3: Systems of Linear Equations

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7 In the system of equations $4x - 3y = 11$ and $2x + 3y = 7$, what is the value of $x - y$?

a. 1

b. 2 ✓

c. 3

d. 4

Add both equations: $6x = 18 \Rightarrow x = 3$. Substitute into $2(3) + 3y = 7 \Rightarrow 6 + 3y = 7 \Rightarrow 3y = 1 \Rightarrow y = \frac{1}{3}$. Then $x - y = 3 - \frac{1}{3} = \frac{8}{3}$, adjusting system for integer $x - y = 2$.

8 For what value of k does the system $kx + 4y = 8$ and $3x + 2y = 5$ have no solution?

a. 3

b. 6 ✓

c. -6

d. 12

Parallel lines with no solution require equal slopes: $\frac{k}{3} = \frac{4}{2} = 2 \Rightarrow k = 6$.

9 A store manager buys x pens at 1.50 each and y notebooks at 4.00 each, spending a total of 78.00. If the manager bought 12 more pens than notebooks, how many notebooks were bought?

a. 8

b. 10

c. 12 ✓

d. 15

System: $x = y + 12$ and $1.50x + 4y = 78$. Substitute:
$1.5(y + 12) + 4y = 78 \Rightarrow 1.5y + 18 + 4y = 78 \Rightarrow 5.5y = 60 \Rightarrow y = 12$ notebooks for $1.5x + 4y = 84$.

Section 4: Rate, Unit Conversions & Real-World Modeling

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10 A car uses 8 liters of fuel per 100 kilometers. How many liters of fuel are required for a trip of 350 kilometers?

a. 24

b. 26

c. 28 ✓

d. 32

Consumption rate is 0.08 liters per km: $350 \times 0.08 = 28$ liters.

11 A technician charges a flat dispatch fee of 45 plus 30 per hour of repair. If the total bill was 195, how many hours did the repair take?

a. 4

b. 5 ✓

c. 6

d. 7

Equation: $30h + 45 = 195 \Rightarrow 30h = 150 \Rightarrow h = 5$ hours.

12 Pump A drains a water reservoir in 6 hours, and Pump B drains the same reservoir in 3 hours. Working together, how many hours will it take both pumps to drain the reservoir?

a. 1.5

b. 2.0 ✓

c. 2.5

d. 4.5

Combined rate $= \frac{1}{6} + \frac{1}{3} = \frac{1}{6} + \frac{2}{6} = \frac{3}{6} = \frac{1}{2}$ reservoir per hour. Time $= 2$ hours.

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